

MYLI JOHNSON

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PROFESSIONAL OVERVIEW

Applied Business Analytics graduate student pursuing doctoral study in Information Systems, with a research background spanning data science, statistics, and business analytics. Experienced in applying statistical and machine learning methods to real-world data, with research interests in anomaly detection, data-driven decision-making, and the application of analytics to business problems. Seeking to build on this foundation through doctoral research and teaching.

EDUCATION

Master of Applied Business Analytics — *Sam M. Walton College of Business, University of Arkansas, Fayetteville, AR*
Aug 2025 – May 2027 (Expected)

GPA: 4.0 / 4.0

Started in the 4+1 program and transitioned to full-time student in May 2026.

Bachelor of Science, Data Science — *College of Engineering, University of Arkansas, Fayetteville, AR*

Aug 2022 – May 2026

GPA: 4.0 / 4.0 | Concentration: Business Data Analytics | Minor: Statistics | Honors Student

Summa Cum Laude, First Ranked Senior Scholar

STANDARDIZED TEST SCORES

GRE General Test (August 2026): Overall Score – 322

Verbal Reasoning — 161 | Quantitative Reasoning — 161 | Analytical Writing — 5.0

HONORS & AWARDS

- Charles B. & Mary E. Friley Endowed Scholarship, Sam M. Walton College of Business (2026–27)
- Russell E. Westmeyer Fellowship in Graduate Studies Scholarship, Sam M. Walton College of Business (2026–27)
- Data Science Practicum Outstanding Team Member Award (2025 – 2026)

RESEARCH

Honors Senior Thesis — Anomaly Identification and Detection Using Statistical and Machine Learning Methods
(2025 – 2026) *University of Arkansas Honors College, Retail Industry Partner*

- Investigated statistical and machine learning methods for detecting anomalies in retail demand forecasting data, aiming to reduce forecast error in perishable food inventory planning by flagging outliers in daily production forecasts.
- Built an end-to-end anomaly detection pipeline on a cloud-based data platform using a continuously growing multi-year dataset, including data cleaning, statistical analysis, and feature engineering segmented by item-store combinations, with anomalies defined as forecast errors beyond ± 3 standard deviations.
- Trained and tuned an Isolation Forest model — selected for its strength on imbalanced data — adjusting classification thresholds to prioritize catching minority-class anomalies, demonstrating a scalable framework combining statistical rules with machine learning.

Data Science Practicum Team Project— Anomaly Detection and AI Alert System

(2025 – 2026) *University of Arkansas Data Science Program, Retail Industry Partner*

- Worked as part of a three-person senior data science team, collaborating directly with a panel of industry experts from the retail partner organization.
- Developed a streamlined anomaly detection alert system for daily fresh-item forecasts for a major warehousing company, training an Isolation Forest model (optimized for class imbalance and to minimize Type-II errors) on operational data via Microsoft Azure Databricks, with feature engineering for temporal trends, holiday effects, and lagged/EWMA variables.
- Built a downstream correction and decision-support pipeline that selects best-estimate replacement values for flagged anomalies and uses a Retrieval-Augmented Generation (RAG) AI agent to explain anomalies and recommend corrective actions to store associates.

JOB EXPERIENCE

Mathnasium Instructor — *Mathnasium, Math Learning Center*

July 2026 – Present

- Instruct and manage a group of students ranging from pre-school through high school age in foundational and advanced math concepts.
- Maintain each student's individualized learning plan, ensuring consistent progress through core math concepts.
- Emphasize conceptual understanding over rote memorization, helping students build the ability to self-direct their learning.

Academic Tutor, Athletics Department — *University of Arkansas*

Aug 2024 – May 2025

- Tutored student-athletes in College Algebra, Pre-Calculus, Calculus I & II, and Introductory Statistics, adapting explanations of difficult concepts to individual learning needs.
- Tracked student progress across sessions to inform ongoing instruction and course outcomes.

Math Tutor — *Arkansas Tech University Math Lab*

Jun 2022 – Jul 2022

- Provided individualized instruction in Calculus, Pre-Calculus, College Algebra, and Statistics.

Private Tutor — *Online Private Tutoring*

Aug 2021 – May 2022

- Designed lesson plans and delivered one-on-one virtual instruction in math, science, and English to a middle-school student.

ADDITIONAL EXPERIENCE

Poll Supervisor — *Johnson County Election Commission*

May 2022 – Jul 2022

- Led a team of poll workers through a primary and runoff election, ensuring compliance with election law and resolving process disputes.

Early Voting Administrator — *Johnson County Clerk's Office*

Jun 2022 – Jul 2022

- Managed early voting equipment and voter communications as part of a four-person team.

TECHNICAL SKILLS

- Python • R • SQL • SAS Enterprise Miner & SAS Viya • SAP S/4HANA • Tableau • Statistical Analysis
- Data Cleaning & Management • Data Visualization • Microsoft Office Suite

RELEVANT COURSEWORK

- Data Management & Databases • Business Analytics • Business Intelligence • AI in Analytics
- Data Visualization & Communication • Optimization Methods in Data Science • Intro to Mathematical Statistics
- Data Structures & Algorithms • Data Science Senior Practicum • ERP Systems • Digital Innovations • Systems Development